

Virtual Brush

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Introduction

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1. USB Camera

2. CSI Camera

Introduction

MediaPipe is an open-source, data-stream processing machine learning application development framework developed and open-sourced by Google. It is a graph-based data processing pipeline used to build applications that consume multiple forms of data sources, such as video, audio, sensor data, and any time-series data. MediaPipe is cross-platform and can run on embedded platforms (such as Raspberry Pi), mobile devices (iOS and Android), workstations, and servers, and it supports mobile GPU acceleration. MediaPipe provides cross-platform, customizable ML solutions for real-time and streaming media.

The core framework of MediaPipe is implemented in C++, with support for languages such as Java and Objective C. The main concepts of MediaPipe include Packet, Stream, Calculator, Graph, and Subgraph.

Features of MediaPipe:

- End-to-end acceleration: Built-in fast ML inference and processing can accelerate even on ordinary hardware.
- Build once, deploy anywhere: Unified solutions work on Android, iOS, desktop/cloud, web, and IoT.
- Ready-to-use solutions: Cutting-edge ML solutions that showcase the full functionality of the framework.
- Free and open source: Framework and solutions under Apache 2.0, fully extensible and customizable.

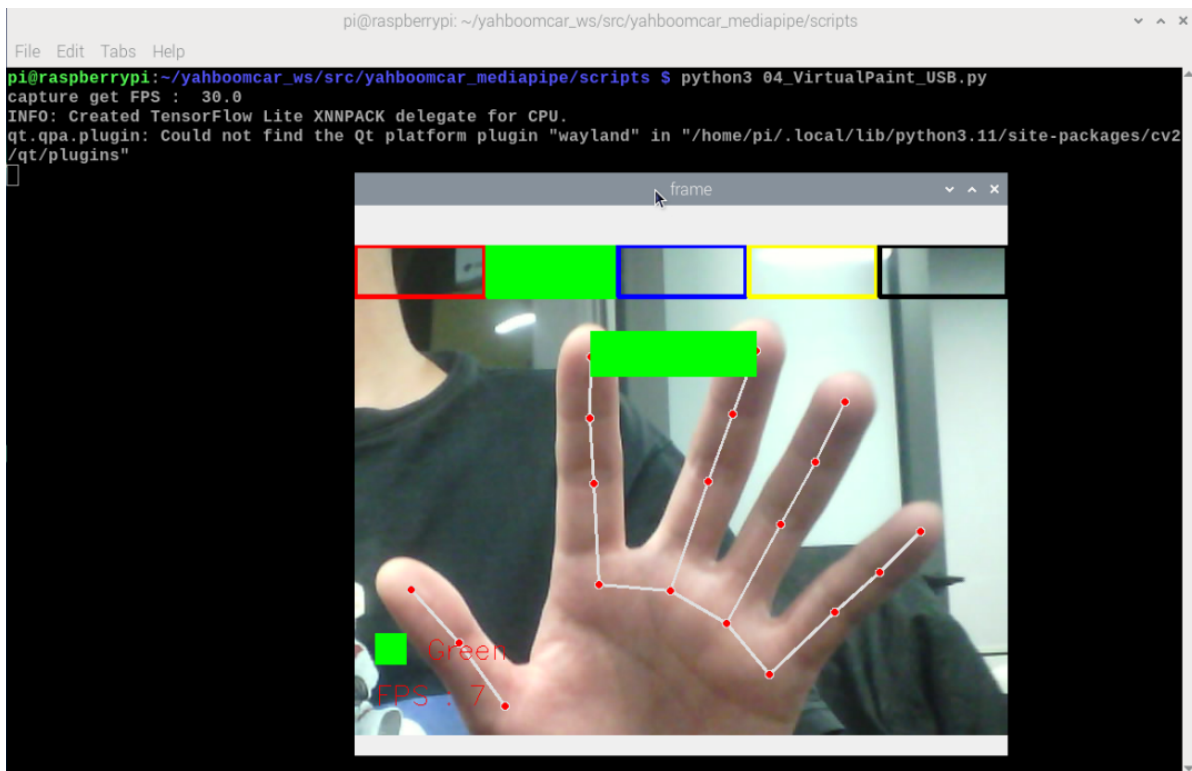
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Source code location: /home/pi/yahboomcar_ws/src/yahboomcar_mediapipe/scripts

when the index finger and middle finger of the right hand come together, it is the selection state, and a color selection box pops up at the same time. When the two fingertips move to the corresponding color position, that color is selected (black is the eraser); when the index finger and middle finger are separated, it is the drawing state, and you can draw freely on the canvas. If you want to exit the program, you can press q in the preview window or press Ctrl+C in the terminal to terminate the program!

1. USB Camera

```
cd /home/pi/yahboomcar_ws/src/yahboomcar_mediapipe/scripts
python3 04_virtualPaint_USB.py
```



2. CSI Camera

```
cd /home/pi/yahboomcar_ws/src/yahboomcar_mediapipe/scripts
python3 04_virtualPaint_CSI.py
```

